Achintya Rai

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Education

University of California, San Diego — Sept. 2024 – June 2027
Bachelor of Science in Computer Engineering — ***GPA: 3.92/4.0***

- Honors & Activities: Competitive Programming Club, NSF REU Fellow (National Science Foundation), UC Scholars Research Fellow
- Coursework: Advanced Data Structures & Algorithms, Machine Learning, Linear Systems, Computer Architecture, Discrete Mathematics

Work Experience

Adobe — San Jose, CA
Incoming Software Engineering Intern — *June 2026 – Sept. 2026*

VALT Health — Remote
Founding Software Engineer — *Aug. 2025 – Feb. 2026*

- Engineered a Flask backend and unified database schemas to ingest **100k+** datapoints across 20+ distinct fitness device sources.
- Implemented asynchronous batch processing in GCP, eliminating pipeline bottlenecks to yield a **35%** throughput increase and **60%** reduction in redundant writes.
- Automated developer environment provisioning via Bash scripts, reducing setup time by **80%**.

Decklar (formerly Roambee) — Sunnyvale, CA
Software Engineering Intern — *June 2024 – Aug. 2024*

- Optimized Python and NumPy data pipelines for Amazon, Coca-Cola, and Samsung, achieving a **15%** capacity boost across **1M+** weekly events.
- Engineered advanced data structures for visualization modules, delivering **20%** faster reporting across internal dashboards.

Research

Graph Theory & Neural Architecture — San Diego, CA
Undergraduate Research Fellow (NSF REU) — *Mar. 2026 – Present*

- Engineered a **Python** pipeline to analyze **LLM** embedding manifolds (Llama-2, DeepSeek-V2), implementing weight-matrix extraction and model export utilities to reduce data footprints from **500GB** checkpoints to **13GB** localized manifolds.
- Profiled and refactored **neural network** placement algorithms and preprocessing routines, achieving a **30x runtime reduction** (30 min → **<1 min**) to enable large-scale inference analysis on resource-constrained **8GB RAM** systems.
- Applied **Rent's Rule** and Louvain Clustering to identify neural network "backbones," discovering noise-filtering increases Inter-Cluster Network density from **0.18% to 13.25%** in Llama-2-70B to guide **model optimization**.

Projects

NanoGPT | *Python, PyTorch, C++, Transformers, LLM Optimization* | — Mar. 2026

- Architected a GPT-style **neural network** from scratch in **Python** and **PyTorch**, hand-implementing multi-headed attention, causal masking, and autoregressive inference logic end-to-end.
- Designed an optimized inference runtime by profiling and compiling performance-critical sub-modules in **C++**, reducing per-token inference latency by **25%** on resource-constrained hardware.
- Parallelized tokenization and data-loading pipelines to maximize throughput, sustaining **100%** GPU hardware utilization across training and inference runs.

Technical Skills

Languages: C/C++, Python, Bash, TypeScript, SQL (Postgres/SQLite), Java, JavaScript
Systems & ML: PyTorch, CUDA, MLIR, LLMs, Quantization, NumPy, Docker, Linux, GitHub Actions
Development: FastAPI, Flask, Django, Asyncio, React, Node.js, AWS, GCP, OpenAI/Gemini API